

Evaluating Limits

ANSWER KEY



Direct substitution and algebraic techniques

1 Evaluate each limit:

a $\lim_{x \rightarrow 1} (3x^2 - 2x + 5) = 6.$

b $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} (x + 2) = 4.$

c $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$ Rationalize: $\frac{(x+4) - 4}{x(\sqrt{x+4} + 2)} = \frac{1}{\sqrt{x+4} + 2} \rightarrow \frac{1}{4}.$

2 Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x^2 - 9}.$

Factor: $\frac{(x-3)(x+2)}{(x-3)(x+3)} = \frac{x+2}{x+3} \rightarrow \frac{5}{6}.$

Trigonometric and special limits

3 Using $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, evaluate:

a $\lim_{x \rightarrow 0} \frac{\sin 5x}{x} = 5 \lim_{x \rightarrow 0} \frac{\sin 5x}{5x} = 5.$

b $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$ (known result: this limit is 0, not ∞).

c $\lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 5x} = \frac{3}{5} \cdot \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \cdot \frac{5x}{\sin 5x} = \frac{3}{5}.$

4 Evaluate $\lim_{x \rightarrow \infty} \frac{5x^3 - 2x + 1}{2x^3 + 7}.$

Equal degrees: the limit is the ratio of leading coefficients, $\frac{5}{2}.$

Limits that fail to exist and the squeeze theorem

5 Explain why $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$ does not exist.

As $x \rightarrow 0$, $\frac{1}{x} \rightarrow \pm \infty$, so $\sin(1/x)$ oscillates between -1 and 1 infinitely often near $x = 0$ rather than approaching a single value.

6 Use the squeeze theorem to evaluate $\lim_{x \rightarrow 0} x^2 \cos\left(\frac{1}{x}\right).$

Since $-x^2 \leq x^2 \cos(1/x) \leq x^2$ and $\lim_{x \rightarrow 0} (\pm x^2) = 0$, the squeeze theorem gives a limit of 0.

Limits with absolute value and piecewise functions

7 Evaluate $\lim_{x \rightarrow 3} |x - 3| \cdot \frac{1}{x}.$

As $x \rightarrow 3$, $|x - 3| \rightarrow 0$ and $\frac{1}{x} \rightarrow \frac{1}{3}$ (a finite value), so the product limit is $0 \cdot \frac{1}{3} = 0.$

- 8 For $f(x) = 2x + 1$ if $x < 3$, and $f(x) = x^2 - 2$ if $x \geq 3$, evaluate $\lim_{x \rightarrow 3^-} f(x)$ and $\lim_{x \rightarrow 3^+} f(x)$, and state whether $\lim_{x \rightarrow 3} f(x)$ exists.
- $\lim_{x \rightarrow 3^-} f(x) = 2(3) + 1 = 7$. $\lim_{x \rightarrow 3^+} f(x) = 3^2 - 2 = 7$. Since both one-sided limits equal 7, $\lim_{x \rightarrow 3} f(x) = 7$ exists.

More algebraic limit techniques

- 9 Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$ by factoring with the difference-of-cubes formula.
- $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$, so the limit is $\lim_{x \rightarrow 2} (x^2 + 2x + 4) = 4 + 4 + 4 = 12$.
- 10 Evaluate $\lim_{h \rightarrow 0} \frac{(2+h)^2 - 4}{h}$.
- $(2 + h)^2 - 4 = 4 + 4h + h^2 - 4 = 4h + h^2 = h(4 + h)$, so the limit is $\lim_{h \rightarrow 0} (4 + h) = 4$.